

O-C plug-in

Overview

This plug-in computes **O-C** (observed minus computed) values for times of light-curve extrema and displays an O-C diagram. A fixed test ephemeris (period and epoch) is compared against observed times of maximum (or minimum) light to reveal timing errors.

Primary reference for the six-clock O-C tutorial (Tables 13.1–13.2): AAVSO *Variable Star Astronomy*, chapter 13 — freely available PDF: [Variable Stars and O–C Diagrams](#) (Grant Foster). The same material appears in Foster, *Analyzing Light Curves*, ch. 13.

Two plug-ins are provided.

Plug-in	Menu	Role
O-C	Tools → O-C...	Main O-C tool (includes Export CSV... in results)
O-C demo data	File → O-C demo data...	Optional Foster clock light curves

The O-C tool can be opened from the **Tools** menu, even when no observations have been loaded, since other than a **From observations** option, an **Imported timings file** option allows a previously exported/created timings file to be opened instead.

Published help (PDF): [O-C on aavso.github.io](#).

Installation

1. In VStar, choose **Tools** → **Plug-in Manager...**
2. Select and install **O-C** (required). Optionally install **O-C demo data** for Foster tutorial light curves.
3. **Close and restart VStar** so new menu items appear.

See also [AAVSO VStar Plug-in Library](#).

Optional: **File** → **Preferences...** → **Plug-in Settings** to check the plug-in download URL matches your VStar version (see the [VStar wiki installation recipes](#)).

Quick start

1. **Tools** → **O-C...**
2. Set **data source**, ephemeris (period and epoch), and other options in the parameter dialog.
3. Click **OK** (a file chooser opens if you chose **Imported timings file**).
4. Review the results dialog: **O-C diagram**, **Data table**, and **Fit summary**.

See **Examples** for Foster demo walkthroughs. **Appendix A** covers all six clocks (demo light curves or imported timings). For a typical workflow on your own AID data (measure → export CSV → re-import), see **Appendix C**.

Examples

These three walkthroughs use the optional **O-C demo data** plug-in (no real star or downloaded timing file required). Install it if needed (see **Installation**), then restart VStar. Each example uses the VSA/Foster test ephemeris **P = 1.0 d**, **epoch = 2450000.0** (Tables 13.1–13.2 in the [Chapter 13 PDF](#)).

Example	Demo scenario (VSA clock)	Purpose
1	Foster clock 4 — epoch jump	Two-segment fit with parallel segments
2	Foster clock 5 — period change	Compare linear, quadratic, and two-segment (different slopes)
3	Foster clock 6 — slowing clock	Compare linear vs quadratic for a curved O-C

Clocks 1–3 (correct ephemeris, epoch offset, period error) are covered in **Appendix A**. Demo menu labels keep the “Foster clock” wording used in the plug-in; the numbers and tables are those of the VSA chapter.

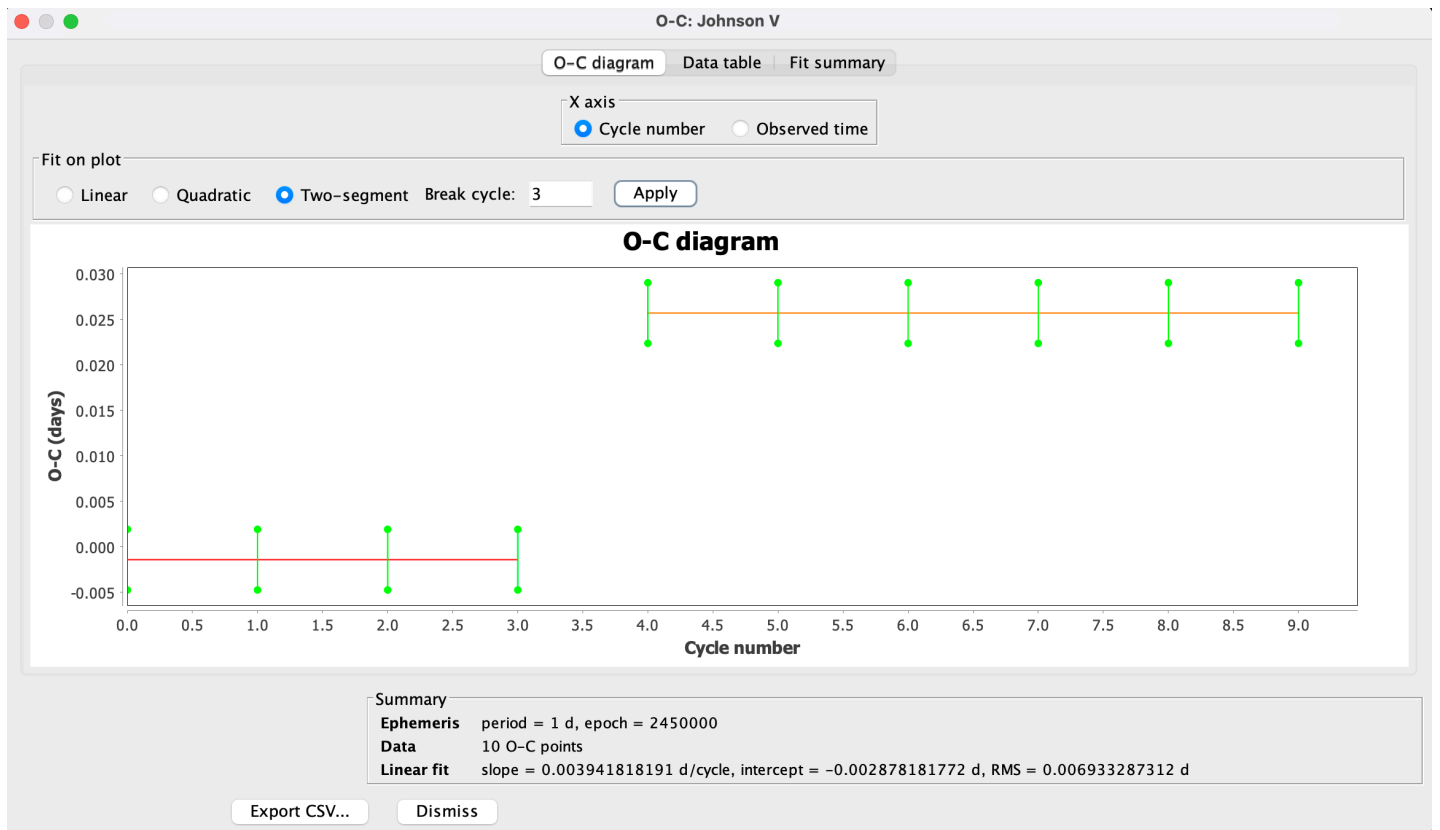
Shared setup for each example (details only in Example 1; abbreviated later):

1. **File** → **O-C demo data...** → choose the demo scenario for that clock → **OK**.
2. **Tools** → **O-C...** → **From observations** → **Star metadata** → **Event**: Maximum → **Timing method**: Parabolic interpolation → **OK** → select **Johnson V**.

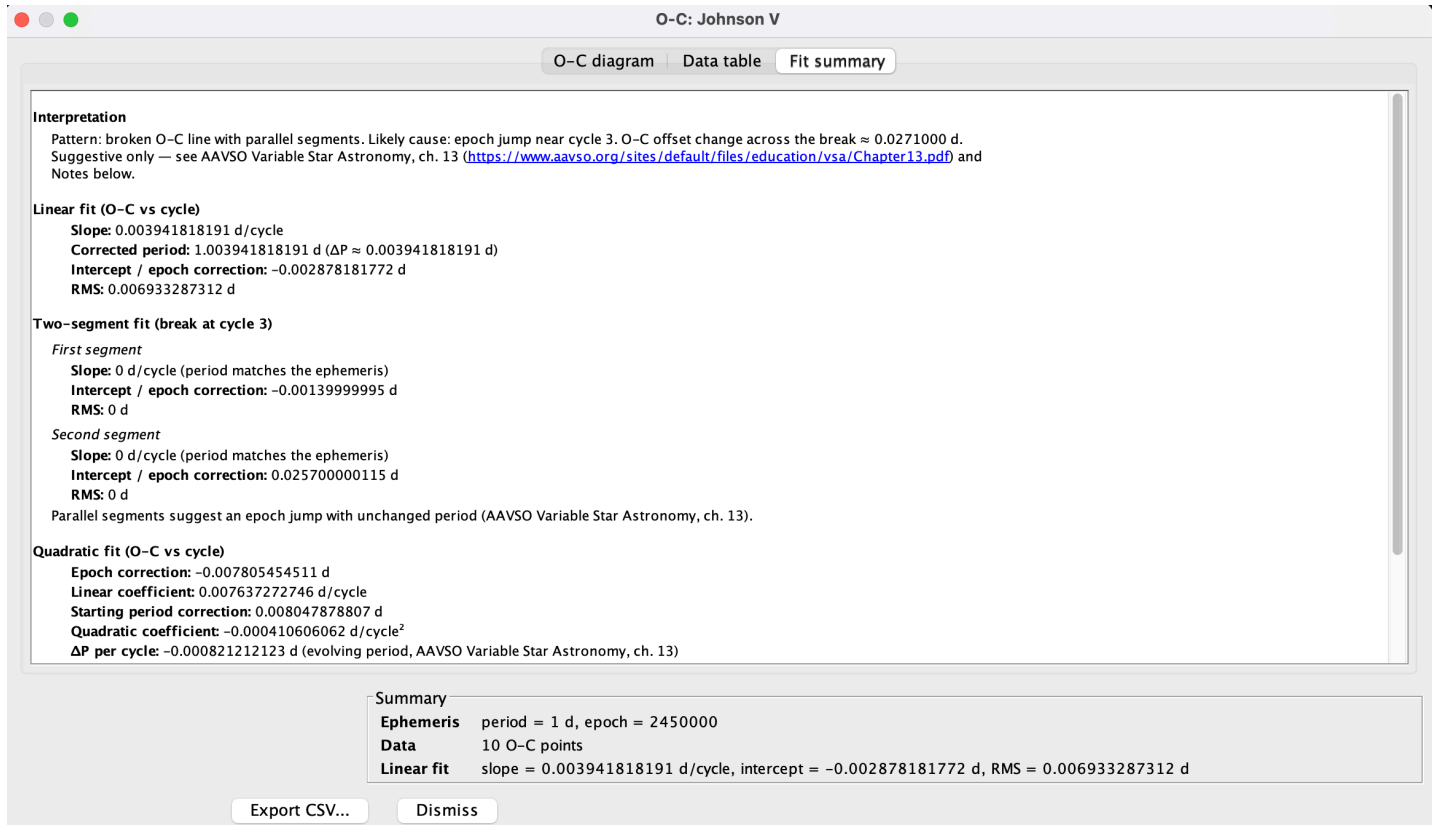
Example 1 — Epoch jump (clock 4)

A sudden step in O-C with unchanged slope on each side (VSA Table 13.2, clock 4).

1. Load **Foster clock 4 — epoch jump** and open O-C as in the shared setup. Note the load-message hint: break cycle **3** (cycles 0–3 are the pre-jump plateau; cycle 4 is already post-jump).
2. On the **O-C diagram**, the points step upward at cycle 4. Select **Two-segment**, enter **Break cycle 3**, click **Apply**. Both segments should be nearly flat (slope ≈ 0) with different intercepts.



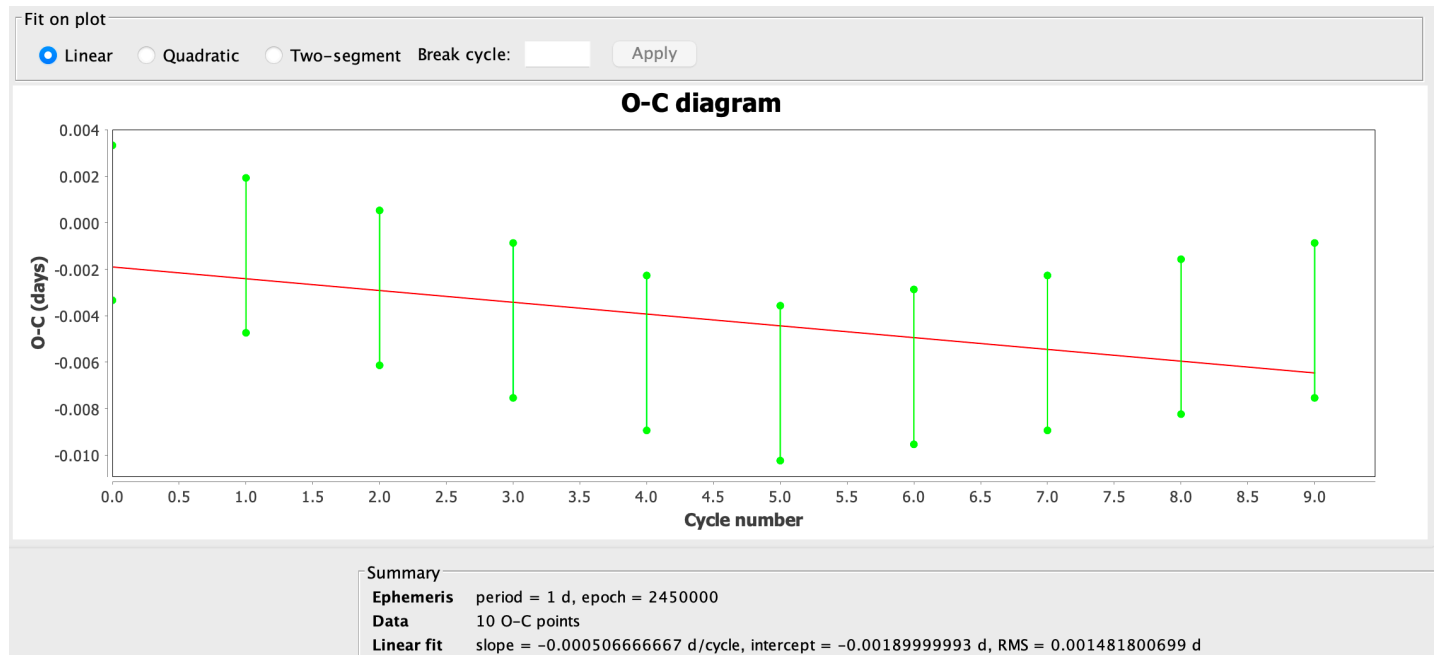
1. Open **Fit summary**. The **Interpretation** should describe parallel segments (epoch jump with unchanged period).



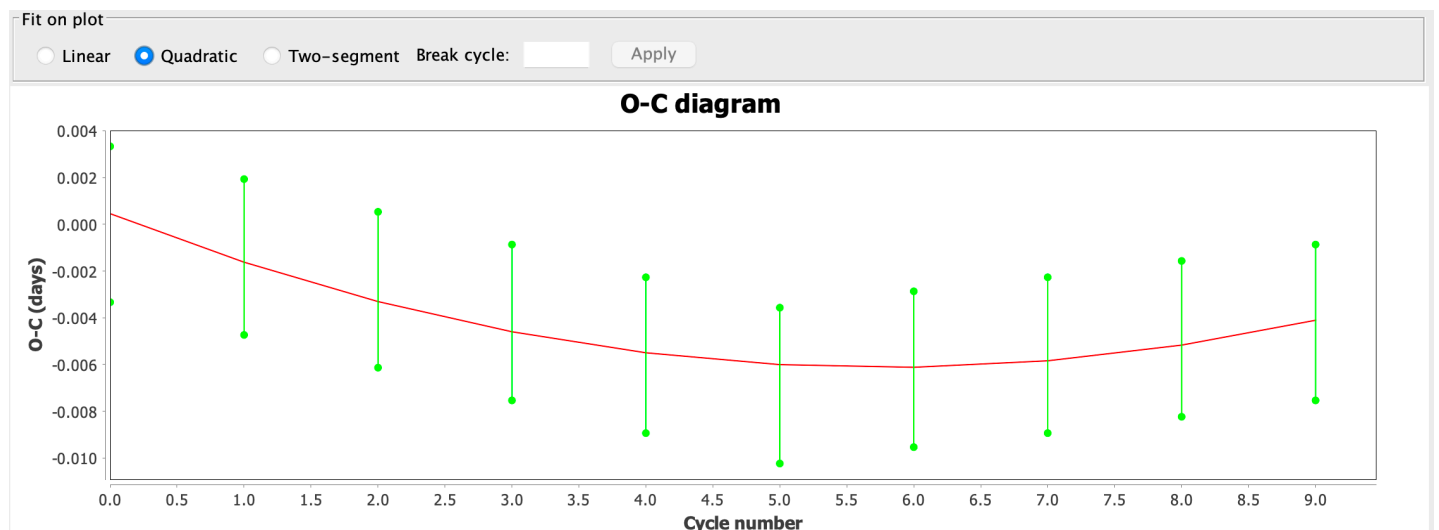
Example 2 — Period change (clock 5)

A slope change mid-run: a single line or parabola is a poor model; two segments with different slopes match Table 13.2 (clock 5).

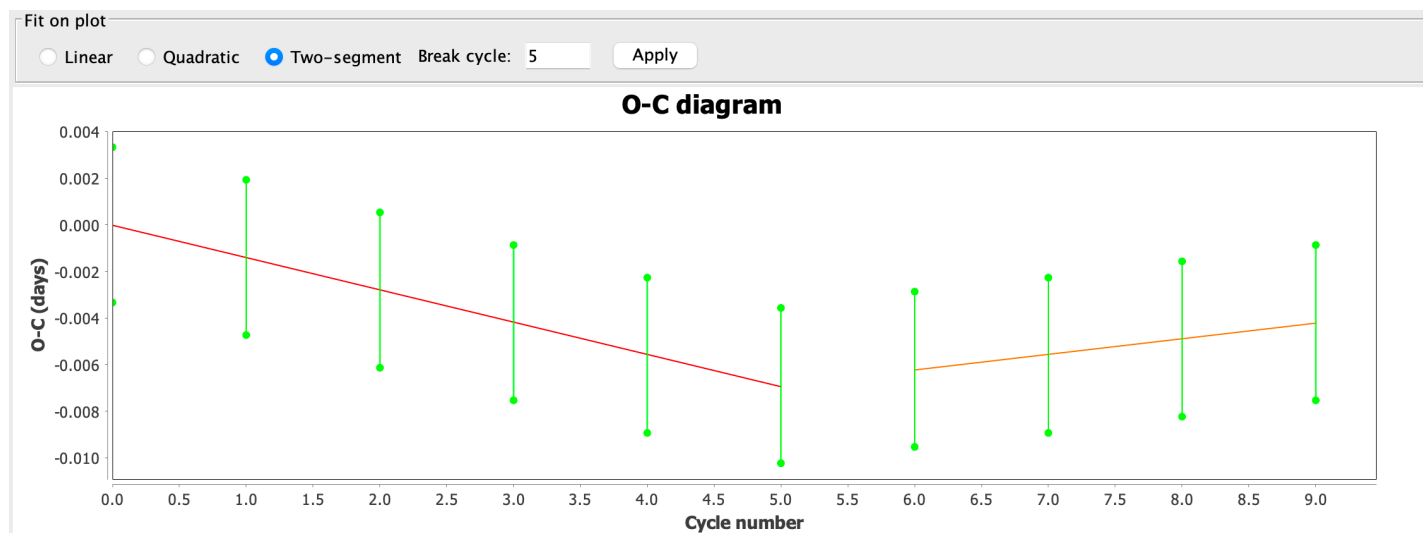
1. Load **Foster clock 5 — period change** and open O-C as in the shared setup. Note the hint: break cycle 5 (cycles 0–5 are the first slope; cycle 6 starts the new regime).
2. Leave **Fit on plot** on **Linear**. The points show a kink; one line is only a compromise across both segments.



1. Switch to **Quadratic**. A parabola may look somewhat better, but it still does not capture a sharp period change at one cycle.



1. Switch to **Two-segment**, enter **Break cycle 5**, click **Apply**. Separate lines before and after the break should show different slopes.



1. Open **Fit summary**. With **Two-segment** selected, the **Interpretation** should report a period change. Switch **Fit on plot** back to **Linear** or **Quadratic** and return to **Fit summary** — the text follows the fit shown on the diagram. Compare with Example 1 (parallel segments / epoch jump).

Interpretation

Pattern: broken O-C line with different slopes. Likely cause: period change near cycle 5. Segment period change ≈ 0.0020557 d/cycle (second minus first segment). Suggestive only — see AAVSO Variable Star Astronomy, ch. 13 (<https://www.aavso.org/sites/default/files/education/vsa/Chapter13.pdf>) and Notes below.

Linear fit (O-C vs cycle)

Slope: -0.000506666667 d/cycle
Corrected period: 0.999493333333 d ($\Delta P \approx -0.000506666667$ d)
Intercept / epoch correction: -0.001899999993 d
RMS: 0.001481800699 d

Two-segment fit (break at cycle 5)

First segment

Slope: -0.00138571426 d/cycle
Corrected period: 0.99861428574 d ($\Delta P \approx -0.00138571426$ d)
Intercept / epoch correction: -0.000019047587 d
RMS: 0.000028171761 d

Second segment

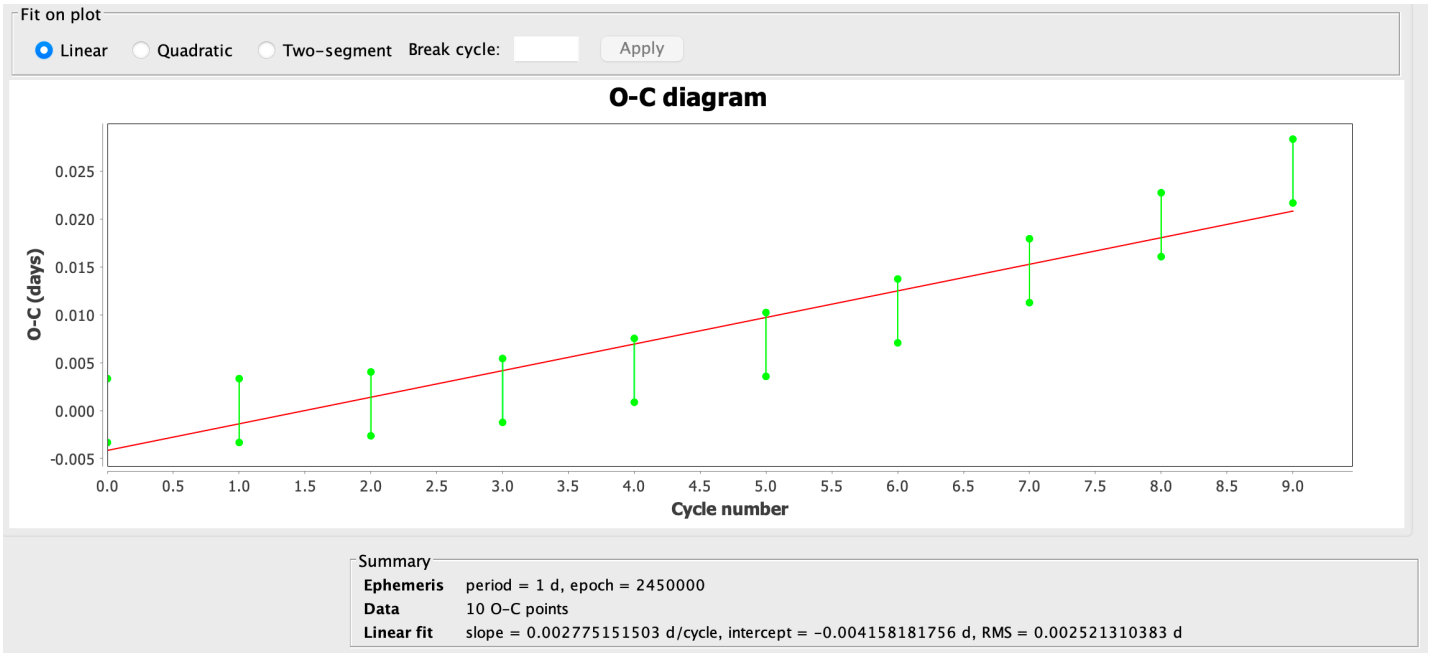
Slope: 0.000670000073 d/cycle
Corrected period: 1.000670000073 d ($\Delta P \approx 0.000670000073$ d)
Intercept / epoch correction: -0.010250000516 d
RMS: 0.000027386039 d

Different slopes suggest a period change near cycle 5 (AAVSO Variable Star Astronomy, ch. 13).

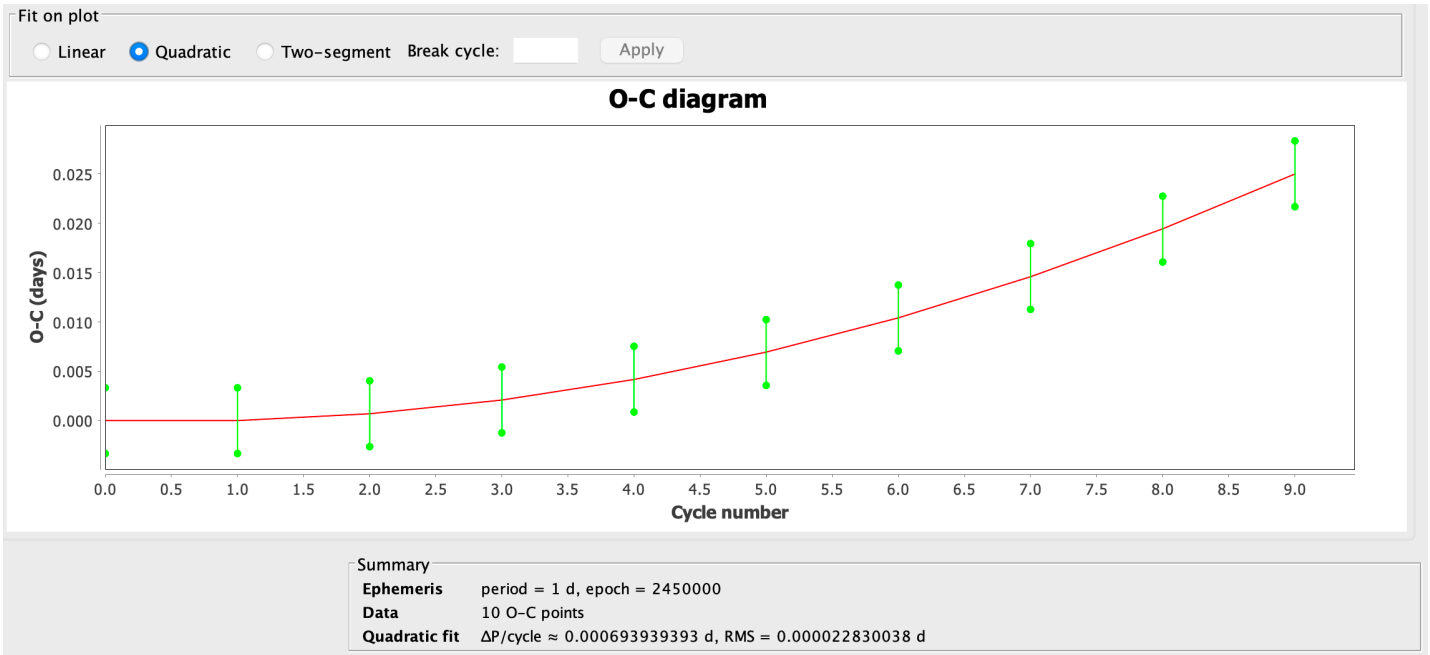
Example 3 — Evolving period (clock 6)

A smoothly curving O-C from an evolving period (Table 13.2, clock 6). Here **linear** vs **quadratic** is the main contrast (no break cycle required).

1. Load **Foster clock 6 — slowing clock** and open O-C as in the shared setup.
2. Leave **Fit on plot** on **Linear**. A straight line cannot follow the curve.



1. Switch to **Quadratic**. The parabola should track the points.



1. **Fit summary** should interpret a curved O-C (evolving period) when **Quadratic** is selected.

Interpretation

Pattern: curved (p
13 (<https://www.aa>

Linear fit (O-C vs c

Slope: 0.00277

Corrected peri

Intercept / epo

RMS: 0.002521

Quadratic fit (O-C \

Epoch correcti

Linear coeffici

Starting perio

Quadratic coe

ΔP per cycle: (

RMS: 0.000022

Notes

Caution (AAVSO \
not white noise —
on O-C autocorre
O-C pattern refer

parabolic) O-C. Likely cause: evolving period. $\Delta P/\text{cycle} \approx 0.0006939 \text{ d}$. Suggestive only — see AAVSO.org/sites/default/files/education/vsa/Chapter13.pdf and Notes below.

ycle)

75151503 d/cycle

iod: 1.002775151503 d ($\Delta P \approx 0.002775151503 \text{ d}$)

och correction: $-0.004158181756 \text{ d}$

1310383 d

vs cycle)

ion: 0.000005454604 d

ient: $-0.000347575768 \text{ d/cycle}$

d correction: $-0.000694545464 \text{ d}$

fficient: $0.000346969697 \text{ d/cycle}^2$

0.000693939393 d (evolving period, AAVSO Variable Star Astronomy, ch. 13)

2830038 d

Variable Star Astronomy, ch. 13): for long-period variables and other stars with cycle-to-cycle period trends may reflect intrinsic period jitter rather than a true ephemeris change. See also: Chapter PDF: <https://www.aavso.org/sites/default/files/education/vsa/Chapter13.pdf>
ence: [AAVSO Variable Star Astronomy, ch. 13](#)

Further reading — X Tri, eclipsing binaries, and other real stars

The VSA chapter also discusses real variables (e.g. X Tri, SU Vir, Z Aur) with O-C diagrams built from published **times of maxima** over many cycles. Those examples need literature ephemerides and timing tables; they are not bundled with this plug-in. The **Examples** above use clocks 4–6 from the [Chapter 13 PDF](#); use **Appendix A** for all six clocks (demo light curves or imported timings). For real stars, obtain timings from the literature or measure extrema from your own observations, then run O-C with **From observations** or **Imported timings** (see **Appendix C** for an AID export/re-import outline).

For **eclipsing binaries**, decades of **eclipse timings** can reveal **apsidal motion** (sinusoidal O-C from general relativity and tidal distortion) or **period change** from mass transfer; those effects need long baselines.

Results dialog

After O-C is computed, a modeless results window opens with three tabs:

Tab	Contents
O-C diagram	O-C points (optional error bars); X axis and Fit on plot controls
Data table	Cycle, observed time, computed time, O-C, uncertainty
Fit summary	Pattern interpretation for the selected fit, plus fit details

Fit on plot — radio buttons choose which fit is drawn:

Option	When available	Notes
Linear	≥2 O-C points	Default; single least-squares line
Quadratic	≥3 O-C points	Curved trend (Foster clock 6 / evolving period)
Two-segment	≥4 O-C points	Enable Break cycle + Apply on the same row

Fit summary — leads with an **Interpretation** paragraph for the fit currently selected on the diagram (VSA / Foster ch. 13 patterns). Fit coefficients and the LPV period-scatter caution follow below.

Export CSV... on the results dialog saves O-C points and fit metadata to a file.

Components (summary)

O-C tool

Parameter dialog uses a compact two-column layout. **Ephemeris source** is a drop-down only (not typable); edit **Period** and **Epoch** in their fields.

After **OK**, choose an observation series or a timings file depending on **Data source**; the results dialog opens next. Use **Export CSV...** there to save results.

O-C demo data (observation source)

Optional. Loads synthetic light curves whose **maximum timings** match Tables 13.1–13.2 of the [VSA Chapter 13 PDF](#) (clocks 1–6). Bump shape is identical for every clock; only timing differs. Then run **Tools** → **O-C...** with **From observations** and the suggested ephemeris from the load message ($P = 1\text{ d}$).

Appendix A — Validation against VSA Tables 13.1–13.2

The six test clocks (theory $P = 1\text{ day}$, **epoch = cycle 0**) are from AAVSO *Variable Star Astronomy* chapter 13 ([PDF](#)). Computed time $C_n = n$. $O-C = O_n - C_n$ (Table 13.2). The same tables appear in Foster, *Analyzing Light Curves*, ch. 13.

Reference O-C values (Table 13.2)

Cycle	Clock 1	Clock 2	Clock 3	Clock 4	Clock 5	Clock 6
0	0	+0.0035	0	−0.0014	0	0
1	0	+0.0035	+0.0021	−0.0014	−0.0014	0
2	0	+0.0035	+0.0042	−0.0014	−0.0028	+0.0007
3	0	+0.0035	+0.0062	−0.0014	−0.0042	+0.0021
4	0	+0.0035	+0.0083	+0.0257	−0.0056	+0.0042
5	0	+0.0035	+0.0104	+0.0257	−0.0069	+0.0069
6	0	+0.0035	+0.0125	+0.0257	−0.0062	+0.0104
7	0	+0.0035	+0.0146	+0.0257	−0.0056	+0.0146
8	0	+0.0035	+0.0167	+0.0257	−0.0049	+0.0194
9	0	+0.0035	+0.0188	+0.0257	−0.0042	+0.0250

Demo scenario mapping

VSA / Foster clock	Demo menu label	Expected in plug-in
1	Foster clock 1 — correct ephemeris	Flat O-C $\approx$ 0
2	Foster clock 2 — epoch offset	Flat O-C $\approx$ +0.0035 d
3	Foster clock 3 — period error	Slope $\approx$ +0.0021 d/cycle
4	Foster clock 4 — epoch jump	Step at cycle 4; break 3
5	Foster clock 5 — period change	Slope change; break 5
6	Foster clock 6 — slowing clock	Curved O-C; select Quadratic on Fit on plot

Method A — imported timings

For each clock, download the timing file (links below), then:

1. **Tools** → **O-C...** → **Imported timings file**.
2. **Period:** 1.0 d, **Epoch:** 2450000.0.

3. Compare O-C and **Fit summary** to Table 13.2.

Method B — demo observation source

See **Examples** for clocks 4–6. For any clock:

1. **File** → **O-C demo data...** → pick a Foster clock (labels match the demo scenario mapping table above).
2. **Tools** → **O-C...** → **From observations**; use the suggested ephemeris from the load message (**Star metadata** pre-fills period and epoch).
3. Compare O-C, **Fit on plot**, and **Fit summary** to Table 13.2.

Timing files (Table 13.1 → times)

Epoch **2450000.0**, period **1.0** d for O-C tool entry. Download from aavso.github.io (published with plug-in docs; same files in `plugin/doc/foster/` in the repo).

Clock	Scenario	Download
1	Correct ephemeris	foster_clock1.txt
2	Epoch offset	foster_clock2.txt
3	Period error	foster_clock3.txt
4	Epoch jump (try break cycle 3)	foster_clock4.txt
5	Period change (try break cycle 5)	foster_clock5.txt
6	Slowing clock (Quadratic on Fit on plot)	foster_clock6.txt

Appendix B — Reference

Ephemeris source

When **From observations** is selected:

UI label	Meaning
Phase plot	Period and epoch from the active phase plot (View → Phase Plot, or Period Analysis → New Phase Plot)
Star metadata	Period and epoch from the loaded star record (often from an AID download or other source that supplies catalogue fields)
Manual entry	Type period and epoch yourself

Disabled for **Imported timings** — enter period and epoch directly, unless the file is an O-C export CSV (metadata is read from the file).

Imported timings file format

One timing per line: `cycle time [sigma]` or `time [sigma]`. Whitespace, comma, or semicolon separators. `#` comments allowed.

Use **one consistent time system** for observations, epoch, and imported timings (JD, HJD, BJD_TDB, etc.). The plug-in does not convert between systems; mix them and O-C will be wrong. Homogenise with VStar's HJD/BJD tools first if needed.

You can also re-open a file saved with **Export CSV...** from a previous O-C run. The tool reads timings from the data rows and, for export files, fills in **Period**, **Epoch**, and **Event** from the `# period=..., epoch=...` comment and the `Event` column (leave those fields blank in the parameter dialog). Export columns are `O_time` and `C_time` (not a specific JD flavour).

Timing methods (From observations)

- **Parabolic interpolation** — parabolic refinement at discrete extremum.
- **Mean JD of extreme N%** — mean time of brightest N% in each cycle.
- **From current model function** — analytic extrema from a JD-based model.

Event type

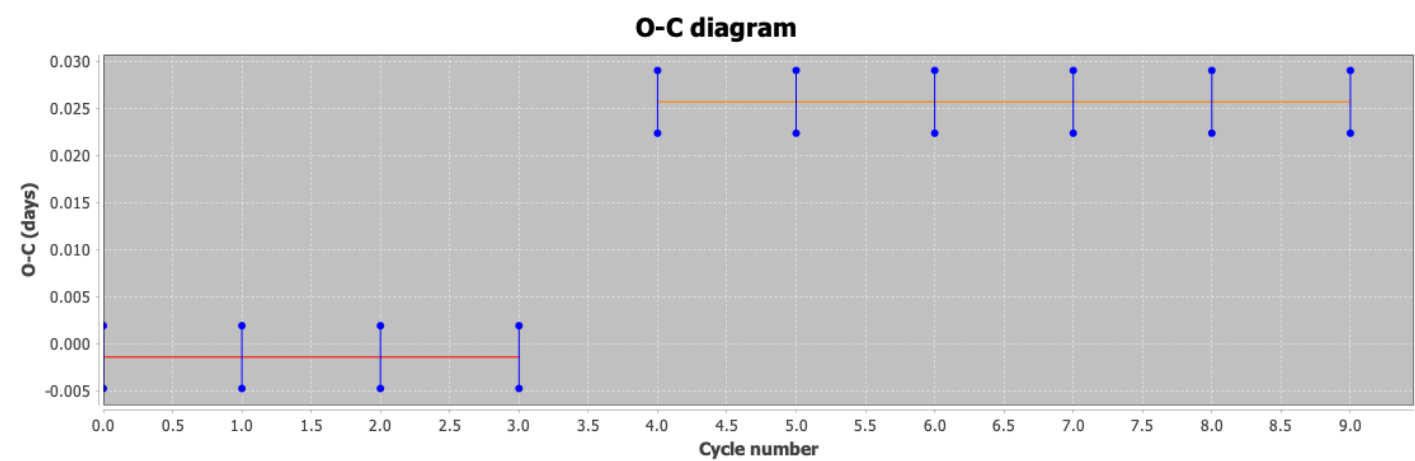
- **Maximum** or **Minimum**
- **Both maxima and minima** — for eclipsing binaries on real data (not used in six-clock tutorial in the VSA Chapter 13 PDF)

Interpreting O-C diagrams

Pattern	Likely cause
Flat at 0	Ephemeris matches
Flat, offset	Epoch wrong, period OK
Linear slope	Period wrong
Broken line, parallel segments	Epoch jump
Broken line, different slopes	Period change
Curved (parabolic)	Evolving period → select Quadratic on the diagram

Two-segment fit

On the **O-C diagram** tab: select **Two-segment**, enter a **break cycle**, click **Apply**. Needs ≥ 4 O-C points and ≥ 2 per segment.



Parameters (initial dialog)

Parameter	Description
Data source	Observations or imported timings
Ephemeris source	Pre-fill (observations only)
Period, epoch	Test ephemeris (required; same time system as the data)
Event	Maximum, minimum, or both
Timing method, extreme N%, min obs	Observations only

Fit summary (after results)

Output	When
Interpretation	Updates with the fit selected under Fit on plot
Linear fit details	≥ 2 O-C points
Quadratic fit details	≥ 3 O-C points
Two-segment fit details	After Apply with a break cycle

Appendix C — AID workflow (export and re-import)

Typical path on your own data: measure extrema from a loaded light curve, export timings, then re-open them without reprocessing the AID.

Suggested target: RZ Cas (AUID `000-BBF-490`), Algol-type EA. Period ≈ 1.20 d (VSX). **Suggested JD range: 2452700 – 2461000** (about 22 years); load **V** only.

1. Load **RZ Cas** from the AAVSO International Database with the JD range and **V** series above.
2. **Tools** → **O-C...** → **From observations** → **Star metadata** → **Event: Minimum** → **OK** → select **V**.
3. Review the diagram, then **Export CSV...** (e.g. `rz_cas_timings.csv`).
4. **Tools** → **O-C...** again → **Imported timings file** → select the CSV. Period, epoch, and event are taken from the export metadata when present.

O-C

Data source:

Ephemeris source:

Period (days):

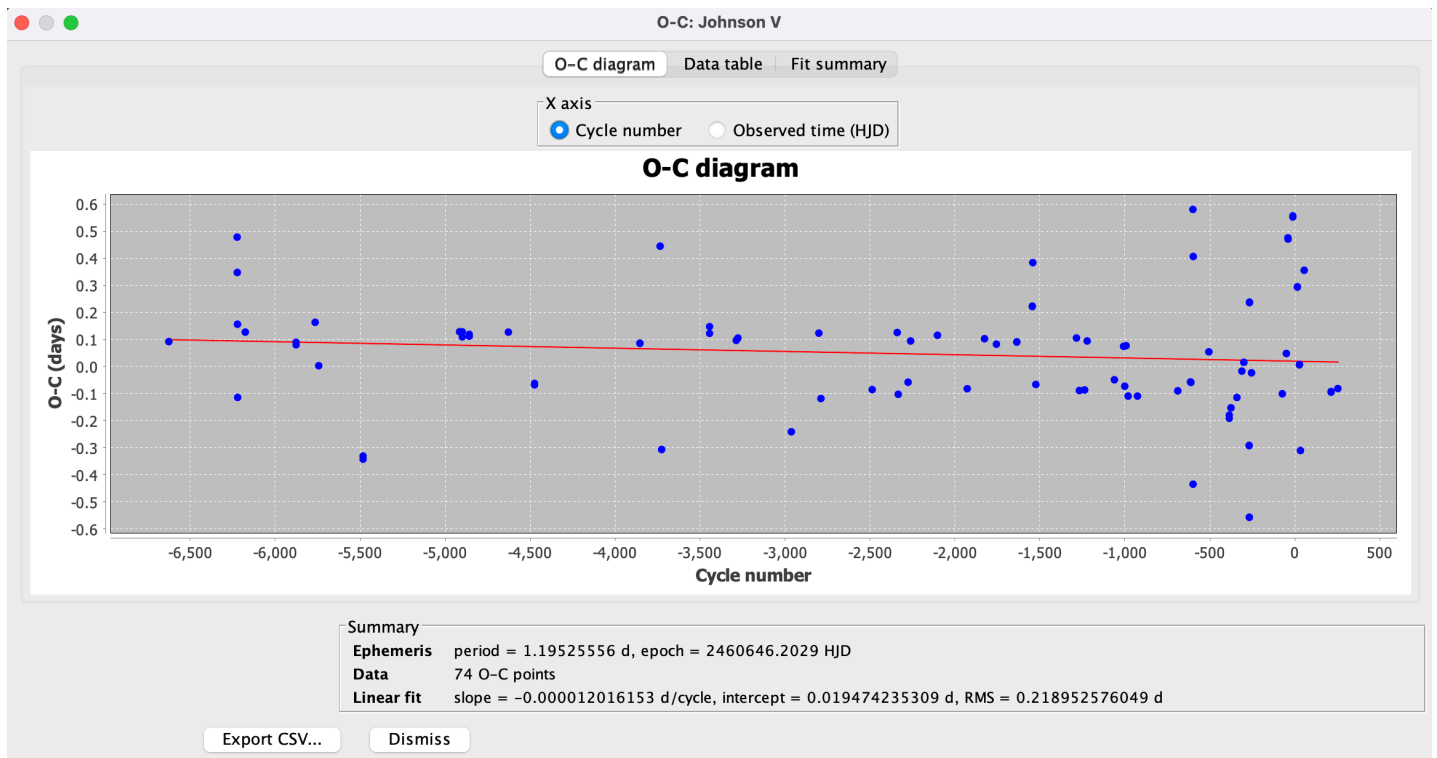
Epoch (HJD):

Event:

Timing method:

Extreme N% (mean timing method):

Minimum observations per cycle:



References

- AAVSO *Variable Star Astronomy*, chapter 13 (Grant Foster) — free PDF: [Variable Stars and O-C Diagrams](#) (primary reference for Tables 13.1–13.2 and the six-clock examples)
- Foster, G. *Analyzing Light Curves*, O-C Analysis, ch. 13, pp. 263–282 (same clock material; book form)
- [VStar plug-in library](#)
- [VStar issue #93](#)

Version history

- **1.10** — Examples rewritten around VSA/Foster clocks 4–6; cite free [VSA Chapter 13 PDF](#) as primary reference for Tables 13.1–13.2; AID export/re-import moved to Appendix C; Results dialog docs updated for **Fit on plot** and interpretation.
- **1.9** — `OAnalysisTool.test()` smoke check (Foster clock 2) for plug-in harness coverage.
- **1.8** — Neutral time labels (`O_time` / `C_time` , Epoch, Observed time); docs stress a consistent time system (JD/HJD/BJD/...) rather than assuming HJD. Removed unused O-C CSV observation-sink plug-in (export is results-dialog only).
- **1.7** — Added screenshots for parameter dialog, results diagram, Fit summary, and two-segment fit.
- **1.6** — User-facing rename: **O-C** (Tools menu) and **O-C demo data** (File menu); plug-in group **Timing**.
- **1.5** — Example 2 uses O-C demo data (Foster clock 2); Foster `.txt` downloads remain in Appendix A Method A only.
- **1.4** — User-facing doc restructure: Plug-in Manager install, examples, appendices, Foster timings on aavso.github.io (absolute URLs), screenshot placeholders.

- **1.3** — Foster Table 13.1 demo timings; six clocks only; compact dialog; general tool; break cycle on Fit summary.
- **1.2** — Imported timings, CSV export, quadratic fit, BOTH event type, demo observation source.
- **1.1** — Model timing, linear/two-segment fits, error bars.
- **1.0** — Initial observation-based O-C plot and table.